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(An Autonomous Institute, Affiliated to Shivaji University, Kolhapur)

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Promoting Excellence in
Teaching, Learning & Research

Department of Computer Science and Engineering

(Artificial Intelligence and Machine Learning) 2026.

THE COMMUNITY ENGAGEMENT PROJECT REPORT ON

"MAHARASHTRA TRANSIT SERVICE"

Smart Bus Tracking & Passenger Management System for Maharashtra

Under the guidance of Miss.Shreya Demanna

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CERTIFICATE

This is to certify that,

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Have successfully completed the project work, entitled

''MAHARASHTRA TRANSIT SERVICE''

In partial fulfilment for the award of degree of Bachelor of Technology in Computer Science and Engineering (Artificial Intelligence and Machine Learning). This is record of their work carried out during academic year 2026.

Date: 07/10/2026

Place: Ichalkaranji

Miss.Shreya Demanna
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DECLARATION

We the undersigned students of S.Y.C.S.E.(AIML) declare that the Community Engagement Project work report entitled “MAHARASHTRA TRANSIT SERVICE” written and submitted under the guidance of Miss.Shreya Demanna is our original work. The empirical findings in this report are based on the data collected by us. The matter assimilated in this report is not reproduction from any readymade report.

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INDEX

CONTENTS	PAGE
1. Introduction	1
2. Problem Statement	2
3. Problem Description	3
4. Facility Requirements Related to Problem	4
5. Field Visit Report	5
6. Technical Solutions on Problem	6
7. Authority Letter	10
8. Conclusion	11
9. References	12

INTRODUCTION

Maharashtra Transit Service is a proposed smart public transportation solution designed to improve the visibility, management, and digital accessibility of the Maharashtra State Road Transport Corporation (MSRTC) bus network. The project focuses on a Smart Bus Tracking & Passenger Management System for Maharashtra. The MSRTC network handles more than 2.5 lakh daily bus trips and approximately 7.5 crore annual passengers, which highlights the need for a reliable digital transit system.

The proposed system aims to bridge the information gap between passengers and the bus network by providing real-time GPS bus tracking, live bus location and estimated arrival time (ETA), digital crew assignment records, online pass and ticket booking, and improved fleet management. Passengers can know where their bus is and when it will arrive, while the transport authority gains a clear digital record of its fleet and crew.

The project follows a six-phase development approach: requirements gathering, system architecture design, bus tracking module development, crew management portal, online ticket and pass booking, and testing, deployment and public rollout.

The technology preferences for the project include GPS hardware with fallback phone-based GPS, Node.js/Python, REST API with WebSocket updates, PostgreSQL and Redis, OpenStreetMap/Google Maps API, React Native, Razorpay/UPI, and NIC/Government Cloud.

In conclusion, Maharashtra Transit Service is more than a tracking application; it is a comprehensive solution that addresses the practical challenges of public bus travel while fostering transparency, efficiency and trust in public transport. Its focus on reliability and accessibility makes it a valuable foundation for a smarter, safer and more connected transit network across Maharashtra.

PROBLEM STATEMENT

Maharashtra's public bus network faces critical gaps in real-time visibility, crew accountability, and digital access for passengers. Surveyed passengers identify unreliable bus information as a major concern, with 86% reporting it. The main problems are:

- No Real-Time Bus Visibility – commuters have no live tracking of bus locations or estimated arrival times.
- No Crew Assignment Tracking – no centralized system tracks driver and conductor assignments on each bus.
- Manual Ticketing Revenue Leakage – manual ticketing and pass issuance leads to long queues and significant revenue leakage.
- Zero Rural Route Monitoring – remote routes in rural Maharashtra have no monitoring or safety checks.
- Unreliable Journey Planning – passengers cannot plan journeys reliably, reducing trust in public transport.

Maharashtra Transit Service addresses these issues through a digital platform that provides live tracking, digital crew records and online booking, making public bus travel faster, more transparent and more reliable.

PROBLEM DESCRIPTION

The project addresses connected passenger and operational problems in the MSRTC public bus network. Passengers need reliable information about bus location and expected arrival time. Without live tracking and ETA information, journey planning becomes difficult and passengers lose time waiting at stops without knowing when a bus will arrive.

The absence of centralized crew assignment tracking makes it difficult to maintain a digital record of driver and conductor deployment. Without such a record, accountability and operational planning remain limited.

Manual ticketing and pass issuance creates queues and administrative workload at counters, and leads to revenue leakage. Rural routes additionally require monitoring and safety visibility, as remote routes currently have no monitoring or safety checks.

Maharashtra Transit Service addresses these problems through an integrated system combining GPS tracking, passenger-facing applications, crew management, online booking, mapping services, databases and government-cloud hosting. This saves passengers' time, improves operational efficiency, and strengthens public trust in the bus network.

FACILITY REQUIREMENTS RELATED TO THE PROBLEM

1. **Real-Time GPS Tracking:** GPS devices and fallback phone-based GPS should provide live bus location through 4G/5G networks.
2. **Passenger Live Location & ETA:** Passengers should be able to view bus location and estimated arrival information through mobile and web interfaces.
3. **Digital Crew Assignment:** A centralized database should maintain driver and conductor assignments for each bus.
4. **Online Ticket & Pass Booking:** Passengers should be able to book tickets and passes online using Razorpay/UPI.
5. **Fleet Management:** Real-time GPS data should support route planning, resource allocation and operational monitoring.
6. **Route Visualization:** OpenStreetMap or Google Maps API should support route and bus-location visualization.
7. **Real-Time Data Infrastructure:** REST APIs, WebSocket updates, PostgreSQL and Redis should support structured and live-location data.
8. **Government Cloud Hosting:** The proposed system should use NIC/Government Cloud infrastructure.
9. **Rural Route Monitoring:** The system should extend visibility and monitoring to remote routes.
10. **Passenger Journey Planning:** Reliable live information should help passengers plan journeys using ETA information.

FIELD VISIT PHOTOS



Image 1 : Visiting Shri Shivtirth Ichalkaranji Madhyavarti Bus Stand (MSRTC, Kolhapur Division) for Community Engagement Program

Field Visit Details

Location: Shri Shivtirth Ichalkaranji Madhyavarti Bus Stand, Ichalkaranji (Maharashtra State Road Transport Corporation, Kolhapur Division)

Purpose: To understand passenger, route-monitoring, crew-management and ticketing requirements relevant to Maharashtra Transit Service.

Team Members Present: Aayaan D. Kalawant, Kunal S. Sapate, Omkar N. Chougule, Yash V. Tare, Vivek M. Chougale.

TECHNICAL SOLUTIONS ON PROBLEM

The proposed technical solution combines hardware, software, databases, mapping, payment services and government-cloud infrastructure to create a connected transit platform. These solutions focus on giving passengers reliable live information while giving the transport authority better control of its fleet, crew and ticketing.

1. GPS Bus Tracking Module

Description:

GPS devices installed on buses provide location data. A phone-based GPS fallback is also proposed so that tracking continues where hardware is unavailable.

Implementation Strategies:

- Install GPS devices on buses to capture live location.
- Use phone-based GPS as a fallback source of location data.
- Transmit location data through 4G/5G networks for live bus tracking.

Benefits:

- Provides live bus location to passengers and operators.
- Forms the base for ETA prediction and fleet monitoring.
- Extends visibility to routes that are currently unmonitored.

2. Backend API and Real-Time Communication

Description:

Backend services connect the buses, the database and the applications, and push live location updates to users as they happen.

Implementation Strategies:

- Use Node.js or Python for backend services.
- Provide system services through REST APIs.
- Use WebSocket updates for real-time location information.

Benefits:

- Delivers location updates to passengers without delay.
- Keeps all modules of the system connected through a single service layer.

3. Database and Live Location Cache

Description:

A database layer stores structured system data, while a cache handles the continuous stream of live-location updates.

Implementation Strategies:

- Use PostgreSQL for structured system data.

- Use Redis for live-location caching.

Benefits:

- Fast access to current bus positions.
- Reliable storage of crew, route and booking data.

4. Passenger Mobile Application

Description:

An Android-first passenger application lets commuters track buses, view arrival information and plan their journeys.

Implementation Strategies:

- Develop the application using React Native.
- Provide live bus tracking and ETA to passengers.
- Support mobile and web interfaces for passenger access.

Benefits:

- Reduces waiting time and uncertainty for passengers.
- Improves journey planning and public trust in the bus service.

5. Maps and Route Visualization

Description:

Map services display routes and the live position of buses so that passengers and operators can see the network at a glance.

Implementation Strategies:

- Use OpenStreetMap or Google Maps API for route and bus-location visualization.
- Display live bus markers along their assigned routes.

Benefits:

- Makes bus locations easy to understand visually.
- Supports route planning and operational monitoring.

6. Digital Crew Management Portal

Description:

A management portal maintains a digital record of driver and conductor assignments for each bus.

Implementation Strategies:

- Maintain a centralized database of crew assignments.
- Link each bus to its assigned driver and conductor.

Benefits:

- 100% digital tracking of driver and conductor deployment per bus.
- Improves accountability and resource allocation.

7. Online Ticket and Pass Booking

Description:

Passengers can book tickets and passes online instead of standing in queues at counters.

Implementation Strategies:

- Integrate Razorpay/UPI for online ticket and pass booking.

Benefits:

- Reduces long queues and manual workload at MSRTC counters.
- Helps reduce revenue leakage from manual ticketing.

8. NIC / Government Cloud Hosting

Description:

The system will be hosted on Maharashtra government cloud infrastructure through NIC to suit a public transport service.

Implementation Strategies:

- Host the platform on NIC/Government Cloud infrastructure.

Benefits:

- Aligns the system with government infrastructure.
- Supports secure and reliable operation of a public service.

9. Testing and Deployment

Description:

The system will undergo comprehensive testing before it reaches the public, followed by a phased rollout.

Implementation Strategies:

- Carry out testing and quality assurance of all modules.
- Run a pilot before the phased public rollout.

Benefits:

- Ensures reliability before public use.
- Allows problems to be corrected during the pilot stage.

10. Future Smart Transit Features

Description:

The platform provides a foundation on which further smart transit capabilities can be built.

Implementation Strategies:

- Crowd monitoring.
- Route optimization.
- Demand forecasting.

Benefits:

- Prepares the system for future growth.
- Supports data-driven decisions about routes and services.

Expected Outcomes

- 30% reduction in passenger wait time through accurate ETA predictions.
- 100% digital tracking of driver and conductor deployment per bus.
- 50% reduction in manual ticketing workload at MSRTC counters.
- Real-time bus location visible to passengers within six months of launch.
- Increased public trust through improved reliability and transparency.
- Foundation for future crowd monitoring, route optimization and demand forecasting.

VISITING LETTER

CONCLUSION

Maharashtra Transit Service is a proposed smart transit solution intended to close the information gap between passengers and the MSRTC bus network. By combining real-time GPS tracking, passenger live-location and ETA services, digital crew tracking, online ticket and pass booking, and fleet-management capabilities, the project provides a structured approach to improving public transportation visibility and operational efficiency.

The proposed technology stack supports the major requirements of the system: GPS hardware and mobile fallback for tracking, Node.js/Python for backend services, PostgreSQL and Redis for data management, mapping APIs for route visualization, React Native for the passenger application, Razorpay/UPI for payments, and NIC/Government Cloud for hosting.

Overall, the project is designed as a foundation for a smarter, safer and more accessible public transit network across Maharashtra, with future scope for crowd monitoring, route optimization and demand forecasting.

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